

AHMED RAYHAN

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Education

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Science in Electrical and Electronic Engineering

CGPA 3.559/4.00

Dhaka, Bangladesh

January 2025

HIGHER SECONDARY SCHOOL CERTIFICATE

Birshrestha Noor Mohammad Public College

GPA 5.0/5.0

Dhaka, Bangladesh

July 2019

Experience

DHAKA POWER DISTRIBUTION COMPANY

Attachment Trainee Intern

Dhaka, Bangladesh

October 2023 – November 2023

- Observed DPDC's organizational structure, including SCADA system, call center, and voltage switching.
- Participated in field visits to understand grid operations and efficient power distribution.
- Gained practical insights into Country's power distribution network.

Research Activities

Performance Analysis of Free Space Optical Communication in Urban Rain Environment

[Accepted Manuscript in IEEE ECCE-2025](#)

This paper investigates the feasibility of Free Space Optical (FSO) communication systems in urban Dhaka, Bangladesh, focusing on the impact of rain on signal performance. It concludes that FSO communication is viable in urban environments with proper mitigation strategies for rain attenuation.

Innovative SPR Biosensor Design for Glucose Monitoring in Blood and Urine

[Under Review, IOP Physica Scripta-2025](#)

In this work, we present a cutting-edge biosensor leveraging surface plasmon resonance (SPR) technology integrated with BaF₂/CdS/Ag/CdS/BP nanomaterials for superior blood glucose detection. The performance was validated through simulations using MATLAB and COMSOL Multiphysics, showcasing minimal error and excellent reproducibility.

One-Dimensional Binary Photonic Crystal for the Detection of Tuberculosis in Blood Samples

[Under Review, Heliyon-2025](#)

This paper presents a novel 1D binary photonic crystal biosensor for detecting Mycobacterium tuberculosis (TB) in blood samples, utilizing refractive index variations to achieve high sensitivity and specificity. The sensor's design, optimized using the transfer matrix method, making it a promising tool for early TB diagnosis.

Leadership and Activities

AUST Career Development Club

Executive Director

Dhaka, Bangladesh

April 2024 – Present

- Manage all IT department functions: strategic planning, project management, and team supervision.
- Lead major event planning and execution, maximizing attendee engagement.

Huawei Seeds For the Future

April 2024-July 2024

- Developed leadership, soft skills, and a winning IoT-based SDG project while networking with industry professionals.

Projects

- Automatic **Voltage Stabilizer**: Built device to regulate AC power output.
- LED Thermometer: Constructed a digital thermometer using **analog ICs**.
- **RFID** Locking System: Designed access control system.
- Gas Leakage Detection: Created **IoT** based system for gas leak detection.
- Implemented **Schematic and Layout** using virtuoso to implement Conventional Fat tree encoder.
- Implemented Home Security System using FSM and executed **Full Chip Design** with cadence Innovus & Genus.

Honors & Awards

- Achieved top 5% academic ranking in Spring '20, Fall '20, and Fall '23 semesters.
- Champion of IEEE's **Research Revelry: EV batteries**
Issued by IEEE AUST STUDENT BRANCH · May 2024
- Champion of IEEE's **Poster Pavilion – A Showcase of Innovation**
Issued by IEEE AUST STUDENT BRANCH · Jun 2023
- 7th Place in **Research Article Writing Competition**
Issued by IEEE AUST STUDENT BRANCH · Aug 2023

Skills & Interests

Technical : MATLAB, C++, Cadence Virtuoso, Innovus, Genus, Proteus, OrCAD PSpice, Arduino.
Soft-skills : Problem Solving, Time Management, Leadership, Adaptability.
Interests : Researching cutting-edge technology & travelling.

References

Dr. Bobby Barua
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Head of the Department
Department of Electrical and Electronic Engineering
Ahsanullah University of Science and Technology
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Hasib Md. Abid Bin Farid
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Head of the Department
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